

An infrared triangle for quantum spin chains

Tobias J. Osborne¹

¹*Leibniz Universität Hannover, Institute of Theoretical Physics, Appelstrasse 2, D-30167 Hannover, Germany*

The infrared triangle of gauge theory—soft theorem, asymptotic symmetry, memory effect—is formulated and proved for quantum spin chains. A truncated symmetry acts only at its endpoints, generating a charge algebra whose central extension is the topological index. A soft magnon decouples with a universal phase slope; granted a scattering hypothesis, a transmitted magnon drags a domain wall by a quantized charge step. The continuum rule that memory is the zero-frequency soft factor fails: the soft factor is a phase, the memory a charge. Bethe and matrix-product numerics verify each claim.

A ferromagnetic spin chain contains two objects that can be pictured without formalism: the *magnon*, a single overturned spin spreading as a wave, and the *domain wall*, the boundary between a region of up spins and a region of down spins. This Letter concerns what each does when the magnon’s momentum is taken small, and the symmetry bookkeeping that controls both. The whole result in one sentence: on the lattice, a soft magnon leaves two-body scattering with a universal phase slope; the symmetry behind this acts at the ends of the chain as a concrete algebra of operators on matrix-product bond space; a magnon that crosses a domain wall displaces it by a quantized charge step—and the continuum rule that identifies the memory with the zero-frequency limit of the soft factor is *false* here, because the lattice triangle closes through charge transport, not phase accumulation.

In continuum field theory, soft theorems, asymptotic symmetries, and memory effects form the *infrared triangle*, three faces of one Ward identity, established for photons and gravitons at null infinity [1, 2]. Hamada and Sugishita extended the triangle to Goldstone bosons of a broken global symmetry [3]; scalar and fracton versions followed [4–6], together with local derivations that avoid null infinity [7] and a careful account of infrared finiteness [8]. Each corner separately has a condensed-matter ancestor: Dyson showed in 1956 that ferromagnetic magnons decouple at long wavelength [9]; Adler zeros of nonrelativistic Goldstone theories are by now systematic [10–13]; and Lan and Xiao described magnon-driven domain-wall displacement as a two-body collision [14, 15]. What has been missing is a lattice statement, at finite spacing, with no continuum limit. We supply one, prove what we can, label the rest conjecture, and report one refutation as a central result.

Setting.—An infinite chain of finite-dimensional sites; a compact symmetry group G acting on-site by $u(g)$; a finite-range G -invariant Hamiltonian H ; and a family of translation-invariant injective matrix-product ground states with tensors A_α and bond dimension χ , one for each vacuum α , permuted by the symmetry as $\alpha \mapsto g \cdot \alpha$. One structural fact is used throughout, the fundamental theorem of MPS symmetries [16, 17]: for an invariant vacuum, acting with $u(g)$ on the physical index of

A_α equals a phase times $V(g)^{-1}A_\alpha V(g)$, where V is a projective representation on the bond space whose class $[\omega] \in H^2(G, U(1))$ is the symmetry-protected topological (SPT) index of the phase [16, 18]. All theorems below are stated in this $(G, \text{injective MPS, finite range})$ register; proofs are linearized in Appendixes A–D, whose ground truth is the repository’s proof shards and machine-checked certificates (Appendix E).

Asymptotic symmetry.—Apply the symmetry to a finite region R only: $U_R(g) = \prod_{x \in R} u_x(g)$. The intertwining relation, used once per site, cancels the V ’s pairwise through the interior, and the whole operation collapses to two bond insertions, $V(g)^{-1}$ on the bond entering R and $V(g)$ on the bond leaving it (Fig. 1). Everything a truncated symmetry does, it does at its two ends. Sending an end to infinity turns this into an asymptotic-symmetry statement, and three facts hold (Appendix A). (i) On states, a half-infinite symmetry string acts *exactly* as a single bond insertion; as an operator sequence it converges strongly if and only if $V(g)$ is a phase. Whenever the symmetry acts nontrivially on the bond space there is no strongly convergent implementer at all—the lattice counterpart of a large gauge transformation. (ii) On finite windows padded around the distinguished bond, the insertions compose as the twisted group algebra $\mathbb{C}_\omega[G]$: the SPT class is precisely the central extension of the asymptotic charge algebra—a group-cohomological multiplier, not a Lie-algebra central charge, since for semisimple symmetries the infinitesimal cocycle gauges away while the torsion class survives. Whether these window states extend to normal states of the physical half-infinite algebra is an open split-property question, hypothesis H-split below; no statement in this Letter uses it silently. (iii) If the symmetry is broken, the half-infinite string leaves the vacuum sector: for every finite region the state is locally a vacuum, and only the limit—in expectation values, never in norm—lands in a *domain-wall* (kink) sector. Kinks are what broken asymptotic symmetries create. When G acts transitively on the vacua, the kink sectors are labelled by ordered vacuum pairs, and the invariant surviving the global symmetry is a double coset in $H_\alpha \backslash G / H_\alpha$, with H_α the unbroken subgroup.

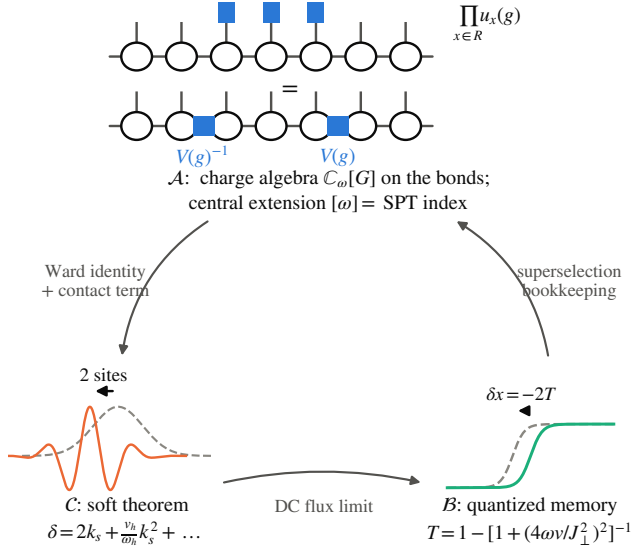


FIG. 1. The infrared triangle at finite lattice spacing. Corner $\mathcal{A}$ (asymptotic symmetry, proved): a symmetry applied to a region of a matrix product state collapses to two bond operators $V(g)^{-1}, V(g)$ at the region's ends; on windows padded about a bond these generate the charge algebra $\mathcal{C}_\omega[G]$, whose central extension $[\omega]$ is the SPT index. Corner $\mathcal{C}$ (soft theorem, proved for two-body scattering): a soft magnon of the ferromagnet exits with phase slope 2, i.e. displaced by two sites. Corner $\mathcal{B}$ (memory, proved conditional on the scattering hypothesis H-AD): a transmitted magnon displaces the wall by exactly $-1/s$ sites, a reflected one by zero. The edge labels name the mechanisms; their present status is uneven and stated honestly in the text: $\mathcal{A} \Rightarrow \mathcal{C}$ holds in the two-body sector and is conjectural beyond it, $\mathcal{C} \Rightarrow \mathcal{B}$ passes soft data to memory only through the transmission probability, and $\mathcal{B} \Rightarrow \mathcal{A}$ is charge bookkeeping within a fixed kink sector.

A dividend of the same telescoping is a lattice Noether identity: on the vacuum, the physical charge density equals the divergence of a purely virtual bond quantity, and for any finite-range invariant H the deformed charge Q_k obeys the exact continuity equation $[H, Q_k] = (e^{ik} - 1)J_k$. One warning, learned the hard way: the kinematic factor $e^{ik} - 1$ is a property of the profile, not a soft factor. The soft theorem has to be earned from the current.

The soft theorem.—Fix the concrete model: the spin- $\frac{1}{2}$ Heisenberg ferromagnet $H = -J \sum_x (\mathbf{S}_x \cdot \mathbf{S}_{x+1} - \frac{1}{4})$. Its polarized ground state is an exact $\chi = 1$ MPS, magnons are one-particle eigenstates with $\omega(k) = J(1 - \cos k)$ and velocity $v(k) = J \sin k$, and $SU(2)$ breaks to $U(1)$, making the magnon a type-B Goldstone mode [10]. Write $S_{12} = e^{i\delta}$ for the exact two-magnon amplitude, δ the continuous phase branch vanishing with the soft momentum k_s , at fixed hard momentum k_h .

Theorem 1 (soft magnon theorem, two-body). *For k_h*

in a compact subset of $(0, \pi)$ and signed $k_s \rightarrow 0$,

$$\delta = 2 \operatorname{sgn}(v_h - v_s) k_s + \frac{|v_h|}{\omega_h} k_s^2 + O(k_s^3) \quad (1)$$

uniformly, with $v_h = v(k_h)$, $\omega_h = \omega(k_h)$.

The zeroth order, $S_{12} \rightarrow 1$, is Dyson's decoupling [9]. The content is the next two orders. The linear coefficient is the pure number 2, up to the sign that records which particle is faster: independent of the hard momentum, of the coupling, of everything. Hard data enters first at second order, through the single even invariant $|v_h|/\omega_h$. And the 2 is not a scattering length—the ferromagnet's scattering length vanishes [13]—it is a *displacement*: the soft packet exits the collision shifted by two lattice sites (Fig. 2). The derivation (Appendix B) does not solve the model. The contact bond imposes one algebraic relation on the outgoing amplitude; the implicit function theorem then fixes the linear coefficient with all hard dependence cancelling. The same relation has a Ward reading: the residue $\langle k_h | Q_0^\dagger J_0^- | k_h \rangle = 2i v_h$ is charge times velocity, the lattice image of the continuum soft-current analysis [3, 11]. Bethe integrability enters only as an oracle: the closed-form expansion reproduces Eq. (1) term by term, and a completeness theorem for the two-magnon sector—proved by direct diagonalization, not assumed—makes the expansion unconditional. Exact diagonalization confirms the form factors to 2×10^{-14} and the two coefficients to 3×10^{-10} ; real-time wavepacket collisions reproduce the displacement reading, the coefficient 2 to 0.2%.

How far beyond two bodies does this reach? Here is the strongest objection we know, at full strength: universality *fails* for unrestricted local sources. There is an explicit four-site operator O_η that creates the same one-magnon state yet shifts the linear soft coefficient by $2i\eta(1 - e^{-3ik_h})$ —a counterexample, not a technicality (Appendix B). What survives is a conditional statement: on the class of Ward-covariant, no-contact sources the soft factor is universal, and we conjecture—Conjecture S—that for n -particle scattering on any injective MPS vacuum in this class, $M_{n+1} = 2ik_s M_n + o(k_s)$ in wavepacket norm. The two-body case is Theorem 1; process independence beyond it rests on three unproved lemmas (wave operators, uniform form-factor regularity, and the soft-limit interchange), stated precisely in Appendix B.

Memory.—Memory needs a wall, so change the model minimally: the easy-axis XXZ ferromagnet, anisotropy $\Delta > 1$, with two product vacua and a domain wall between them. Send in a magnon wavepacket and ask where the wall sits afterwards (Fig. 3). Read the wall position from a window W around it, $\mathfrak{X}_W = \text{const} + \frac{1}{2s} \sum_{x \in W} S_x^z$. Its change over the event is an exact operator identity, no hypotheses: the time-integrated $U(1)$ current through the two boundary bonds of W , i.e. the zero-frequency

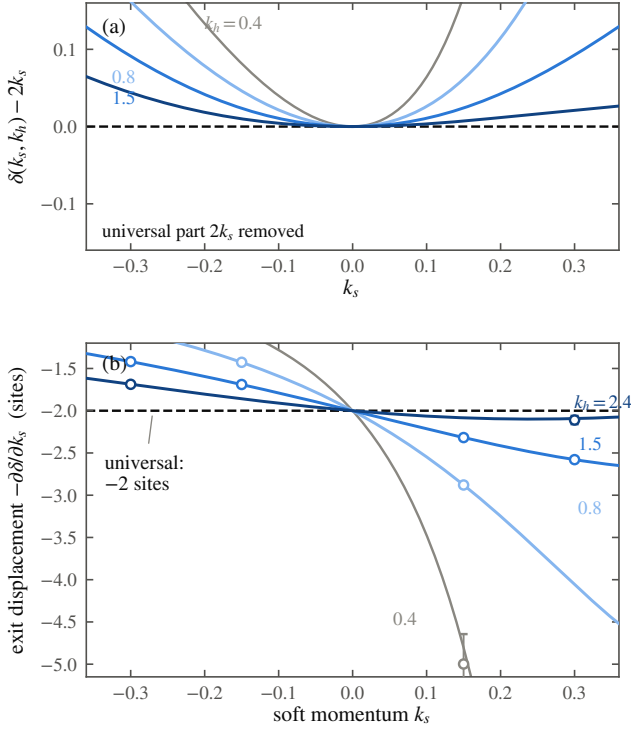


FIG. 2. The soft magnon theorem and its verification. (a) The exact two-magnon phase of the Heisenberg ferromagnet with its universal part removed: $\delta(k_s, k_h) - 2k_s$ vanishes quadratically for every hard momentum k_h —the hard leg enters only at order k_s^2 , with coefficient v_h/ω_h , Eq. (1); exact diagonalization confirms both coefficients to 3×10^{-10} . (b) The displacement reading: the soft packet’s exit displacement $-\partial\delta/\partial k_s$ for all k_h (curves, exact) passes through -2 sites at $k_s = 0$. Circles: real-time wavepacket measurements, which reproduce the universal coefficient 2 to 0.2%. The gray curve ($k_h = 0.4$) shows the non-uniformity of the expansion as $k_h \rightarrow 0$; the universal -2 survives it. Every soft magnon leaves a two-site footprint, whatever magnon it scatters off—for what it takes to drop the last qualifier, see the text.

weight of the physical boundary current. Memory is the DC component of a current at the edge of the observation window—the lattice image of the continuum statement that memory sits in the field at the boundary [2, 7]. Charge bookkeeping then gives the memory law, in the general register.

Theorem 2 (memory law). *Let G be compact, $\{A_\alpha\}$ a covariant family of injective MPS vacua of any bond dimension, H finite range and G -invariant, and fix a kink sector between two vacua carrying charge density $\pm s$ along a common unbroken $U(1)$. Assume the scattering hypothesis H-AD (Appendix C): wave operators exist for the kink-magnon event, with one reflected and one transmitted channel of definite relative charges ∓ 1 , and local*

decay. Then the wall-displacement operator is

$$\Delta X = -\frac{1}{s} N_T, \quad \delta x = -\frac{1}{s} \langle N_T \rangle, \quad (2)$$

with N_T the transmitted-channel projection: each transmitted magnon displaces the wall by exactly $-1/s$ sites, each reflected one by zero, and $\text{spec}(\Delta X) \subseteq \{-1/s, 0\}$ with Bernoulli variance.

The displacement is quantized by charge conservation alone: it does not depend on momentum, anisotropy, packet shape, or any scattering phase; those enter only through $\langle N_T \rangle$. The proof is bookkeeping—a transmission changes the separated leg charge by 2, and the wall must absorb it at $2s$ per site—but one hypothesis is structural: the two vacua must carry *opposite* charge density along the same unbroken direction, which forces the broken element to invert that direction, as the spin flip does in XXZ; a bare abelian symmetry cannot produce such a kink pair. H-AD itself we prove only for the dominant domain-wall sector of the XXZ chain, conditional on an explicitly stated sector-reduction hypothesis; for the full chain it remains an assumption (Appendix C). Sparse-sector numerics on chains of 160 sites confirm $\delta x = -2 \langle N_T \rangle$ at $s = \frac{1}{2}$ to 0.004 sites.

In the reduced sector the kink-magnon problem is a side-coupled (Fano) level, giving the closed form $t(k) = [1 + iJ^2/4\omega(k)v(k)]^{-1}$ with $\omega(k) = J(\Delta - \cos k)$; the measured reflection matches this over $\Delta \in [1.5, 12]$ to 0.9–5.8% with no fitted parameter (criterion 8%, fixed in advance). As $k \rightarrow 0$,

$$T(k) = 16(\Delta - 1)^2 k^2 + O(k^4), \quad (3)$$

so a soft magnon is totally reflected and leaves no memory: the memory effect has an Adler zero of its own. Whether the zero’s coefficient is universal—a function of the kink’s asymptotic data alone—we have not established; that remains a conjecture.

Now the refutation. The continuum triangle suggests a dictionary: memory = DC limit of the soft factor, summed over the event. That statement is *false* in the XXZ chain, twice over: the displacement per transmission is fixed by charge conservation whatever the scattering phase does, and a purely transmitting wall with identically zero phase shift still displaces by exactly $-1/s$. The soft factor is a phase; the memory quantum is a charge. What survives—and is proved—is the pair above: memory is the DC weight of the boundary current, and soft data reaches it only through the transmission probability. We regard this not as a failure of the triangle but as its lattice sharpening: the third corner closes through charge transport, not phase accumulation.

The remaining edge closes at the level of bookkeeping, and only there: under finite-range dynamics the vacuum-pair label of the kink sector is rigid at finite times, and the displacement obeys $2s\delta x + (q_{\text{out}} - q_{\text{in}}) = 0$ within that

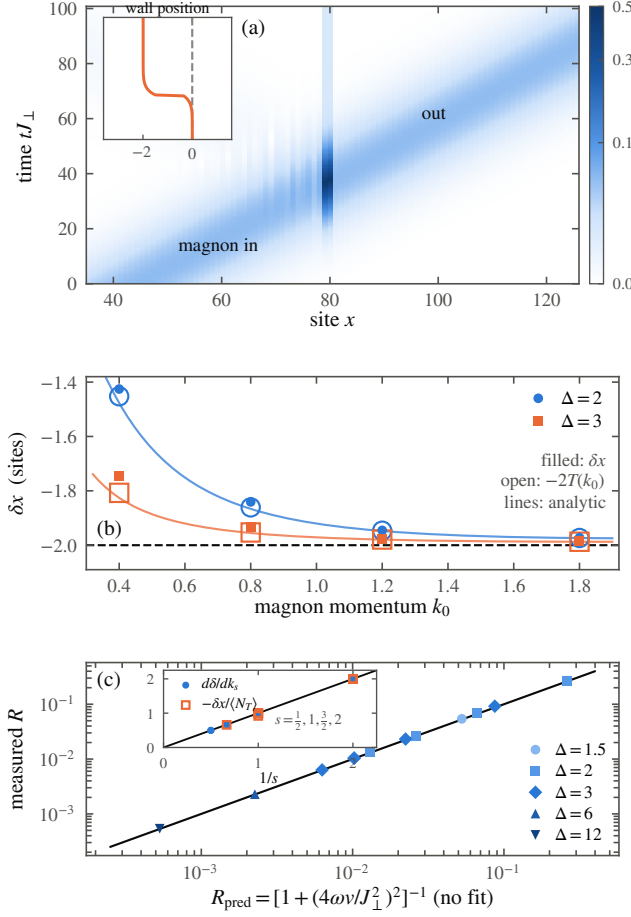


FIG. 3. The quantized memory effect in the XXZ chain ($\Delta > 1$). (a) Space-time record of one event ($\Delta = 3$, $k_0 = 1.2$, $N = 160$): flipped-spin density as a magnon wavepacket crosses the wall. Inset: the wall position, which steps back by two sites during the collision and stays there—the permanent record. Across the scan the displacement obeys $\delta x = -2 \langle N_T \rangle$ to 0.004 sites. (b) Displacement versus momentum: filled symbols δx , open symbols $-2 \langle N_T \rangle$, curves the parameter-free Fano prediction. (c) The measured reflection collapses onto $R = [1 + (4\omega v/J^2)^2]^{-1}$ across $\Delta \in [1.5, 12]$, $k_0 \in [0.4, 1.8]$, to 0.9–5.8%; as $k_0 \rightarrow 0$ transmission vanishes quadratically, Eq. (3): a soft magnon leaves no memory. Inset: the spin- s falsifier of Conjecture B. Soft phase slope (filled, rings up to 480 sites) and memory quantum $-\delta x / \langle N_T \rangle$ (open, wavepacket runs) against the single line $1/s$, for $s = \frac{1}{2}, 1, \frac{3}{2}, 2$ (memory: $s \leq \frac{3}{2}$). Both follow the charge datum, not the constant 2.

fixed sector. That the memory *reconstructs* the asymptotic transformation which acted remains open.

One invariant, read out twice.—Two 2’s appear above: the soft phase slope of Theorem 1 and the spin- $\frac{1}{2}$ memory quantum $1/s$ of Theorem 2. We conjecture—Conjecture B—that both equal $|q_{\text{hard}}|/s$, the hard particle’s $U(1)$ charge in units of the vacuum spin density: one asymptotic-charge datum read out as a phase in cor-

ner $\mathcal{C}$ and as a displacement in corner $\mathcal{B}$. The conjecture was built to fail at $s \neq \frac{1}{2}$, where the candidate laws $1/s$ and the constant 2 separate; it survived (Fig. 3c, inset). Two independent implementations measured the phase slope $= 1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ and the memory quantum $= -1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}\}$, within pre-registered 8% bands (measured deviations $\leq 1\%$ at clean kinematics); an exact two-magnon contact computation gives the slope $1/s$ in closed form, with all hard dependence cancelling. The $|q_{\text{hard}}|$ factor is untested: every leg so far carries $|q| = 1$, and a charge-2 bound-state projectile is the next falsifier.

Where the SPT index lives.—Corner $\mathcal{A}$ made $[\omega]$ the central extension of the soft charge algebra; does it show up in soft amplitudes? The answer is a rigidity dichotomy, and it is proved (Appendix D). In the *bulk*, a closed symmetry insertion has two endpoints whose projective multipliers cancel exactly, and every normalized soft coefficient is a continuous function along symmetric gapped deformation paths: nothing bulk is a topological invariant without a separate local-constancy proof. Along a symmetric path through the AKLT phase, a natural bulk soft-charge coefficient moves from 0.125 to 0.240 with the SPT class fixed. At an *unpaired endpoint* the situation inverts: the compensated soft residue is an operator on the χ -dimensional edge register whose centered charge has spectrum in the $[\omega]$ -shifted lattice $q_\omega^\circ + \mathbb{Z}$ and dimension at least that of the smallest ω -projective representation. For the AKLT half-chain the residue is exactly $-\frac{1}{2}[1 - (2b^2 - 1)^L]Z \rightarrow -Z/2$ along the entire path: eigenvalues $\pm \frac{1}{2}$, rigid, while the bulk coefficient drifts. The topological data of soft physics concentrates at the boundary. Physical edge statements carry the hypothesis H-split above, and charge bookkeeping then quantizes edge-memory outcomes exactly as in Theorem 2: what is protected is the memory’s *capacity*—a projective multiplet no symmetric perturbation can split—not any particular amplitude. That an AKLT boundary magnon actually writes to it, we state as a conjecture with a definite Hamiltonian and a pre-registered test (Appendix D).

Outlook.—The topological data of the chain thus concentrates in a boundary soft algebra, $\mathbb{C}_\omega[G]$ at the ends, in structural parallel to the celestial program’s discovery of symmetry towers at the boundary of spacetime [1]. The symplectic origin of this charge algebra—the moment-map geometry of the MPS manifold behind the bond operators—is deferred to forthcoming work. The asymmetry of our opening has meanwhile dissolved: the wall is a broken asymptotic symmetry made visible, and the magnon’s soft phase and the wall’s memory measure one conserved charge crossing an observer. The infrared structure of gauge theory and gravity, long tied to null infinity, fits on a chain of spins, where every step of it can be checked.

We thank D. Giulini for discussions.

Corner A: the asymptotic charge algebra

Ground truth: repository shards
theory/corner-a.md, corner-a-kinks.md,
corner-a-goldstone.md (claim rows WI, A1, A2, G0,
all PROVED through the adversarial verification loop
recorded there); numerical certificates in Appendix E.
Throughout, α labels vacua, $H_\alpha = \{g : g \cdot \alpha = \alpha\}$ the
unbroken subgroup, and phases are normal-ordered so
that the extensive phase of the truncated symmetry is
removed.

Truncated symmetry (WI). For a finite region $R = [a, b]$ and any $g \in G$, exactly and with no limit,
 $U_R(g) |\psi(A_\alpha)\rangle = e^{i|R|\theta_\alpha(g)} |\psi(T_R^{(g)})\rangle$, where $T_R^{(g)}$ carries
 A_α outside R , $A_{g \cdot \alpha}$ inside, and two bond insertions:
 $V_\alpha(g)^{-1}$ on $\partial_- R$ and $V_\alpha(g)$ on $\partial_+ R$. The window must
contain one site beyond each cut (or carry explicit edge-
bond insertions); with interior bonds only, the identity is
false, and the orientation of the two insertions matters—
both failure modes are exhibited numerically in the cer-
tificate.

Unbroken case (A1). (a) Half-infinite strings act on
states exactly as single bond insertions, and the conver-
gence is eventually constant on every local algebra, not
asymptotic. (b) They are implemented by a strongly con-
vergent operator sequence if and only if $V_\alpha(g)$ is scalar.
(c) Two insertions give the same state iff they differ
by a scalar, so the endpoint states form a $PGL(\chi)$ -
torsor. (d) On windows padded about the bond the
map $M \mapsto$ insertion is injective, and the insertions re-
alize a linear representation of the twisted group algebra
 $\mathbb{C}_{\omega_\alpha}[G]$ in which the multiplier acts nontrivially; padding
is necessary—an explicit $\mathbb{Z}_2$ -symmetric counterexample
shows the unpadded action is ill defined. On states the
multiplier dies and what remains is a homomorphism
 $G \rightarrow PGL(\chi)$; what $[\omega_\alpha]$ obstructs is lifting it to an
honest unitary representation. (e) *Open (H-split):* that
these endpoint states are normal states of the physical
half-infinite algebra—equivalently that $\mathbb{C}_\omega[G]$ acts on a
physical edge Hilbert space—is not proved; the expected
route is the split property. Every physical-edge statement
in this Letter carries this hypothesis explicitly.

Broken case (A2). For $g \notin H_\alpha$, every finite trunca-
tion stays in the vacuum sector (a kink-antikink pair at
the two ends); the half-infinite limit exists in the weak-
* sense only, with an exponential rate, and lands in the
kink sector $\mathcal{K}_{\alpha, g \cdot \alpha}$, disjoint from the vacuum folium. The
jump is invisible at every finite truncation and created
entirely by the surviving end. Under the transitivity hy-
pothesis (T), vacuum pairs form $(G/H_\alpha) \times (G/H_\alpha)$ and
the complete diagonal invariant is the double coset in
 $H_\alpha \backslash G/H_\alpha$; for $G = SU(2)$ broken to $U(1)$ this is the
relative angle of the two vacuum axes. Without (T) the
classification holds per orbit. Two load-bearing gaps are
recorded in the shards and respected here: H-split above,

and the absence of a g -uniform statement when the vac-
uum manifold is continuous.

Currents (G0). For any generator ξ and finite-range
invariant H : $[H, q_x(\xi)] = j_{x-1|x}(\xi) - j_{x|x+1}(\xi)$ with the
local cut current, and in wavepacket form $[H, Q_k] =$
 $(e^{ik} - 1)J_k$. On the vacuum, the charge density acts as
the lattice divergence of a virtual bond quantity—the
potential is the fundamental object, as a theorem. The
finite-window Fourier identity carries two $\Theta(1)$ boundary
terms; the clean $(1 - e^{ik})$ form holds only in the stated δ -
normalized sense. The kinematic factor supplies no soft
theorem (a withdrawn overclaim is recorded as a refuted
row in the claims ledger).

Corner C: the two-body soft theorem and its limits

Ground truth: theory/soft-current-recon.md (row
S2-2body, PROVED), theory/oracle-bethe.md
(oracle facts O1–O10, verification role only),
theory/ml2-completeness.md (row ML2,
PROVED), theory/ml4-ward-reduction.md and
theory/ml5-universality.md (rows ML4-A, ML4-
Ward, ML5-A/B PROVED as conditional implications;
ML5 unrestricted REFUTED).

Derivation of Theorem 1. The exact current of the
model is $j_{x,x+1}^- = \frac{J}{2}(S_{x+1}^- - S_x^-)P_{x,x+1}$. A two-magnon
eigenwave is free away from contact; writing the incoming
coefficient as 1 and the outgoing as $s(k_s, k_h)$, the single
contact bond imposes

$$(2z_h - z_s z_h - 1)s + (2z_s - z_s z_h - 1) = 0, \quad z = e^{ik}, \quad (4)$$

whose unique analytic solution near $k_s = 0$ has k_s -
coefficient $2i$, all hard dependence cancelling; the contin-
uous logarithm gives Eq. (1), with remainder uniform on
compact hard sets. The Ward reading: the state $Q_0 |k_h\rangle$
is the Bethe descendant with one rapidity at infinity, and
 $\langle k_h | Q_0^\dagger J_0^- | k_h \rangle = 2iv_h$; dividing a hard external-leg reduc-
tion by the energy shift $v_h k_s$ cancels the velocity and
leaves the 2. Symmetry alone does not prove the theo-
rem: the current has an orthogonal contact component,
and what makes it subleading in the two-body sector
is energy-shell matching plus C^1 regularity of the on-
shell trace (an abstract cancellation lemma proved in the
shard), not the Ward identity. Completeness of the two-
magnon resolution—scattering states, one bound band,
descendants, and one singular vector on even rings—is
proved by direct Jacobi diagonalization; no Bethe com-
pleteness is assumed, and integrability appears nowhere
in the hypotheses.

The universality boundary. With the four-site local op-
erator $D = S_0^- S_1^- - S_1^- S_2^- + S_2^- S_3^- - S_0^- S_3^-$, the source
 $O_\eta = S_0^- + \eta D$ creates the same one-magnon ampli-
tude as S_0^- but shifts the linear soft coefficient of the
induced two-body amplitude by $2i\eta(1 - e^{-3ik_h})$. Unre-
stricted process independence is therefore refuted. The

repaired statement (proved as a conditional implication) holds on the source class $\mathcal{S}_W$ defined by five conditions—vanishing soft intercept and first contact jet, second-derivative norm control, an exhaustive LSZ decomposition, and reduced-channel matching—and Conjecture S asserts universality on exactly this class. Open, and stated as such: infinite-volume wave operators (ML1), uniform form-factor regularity including the $k = \Theta(1/N)$ regime (ML3), the $n \geq 2$ -hard channel (ML4 beyond one hard leg; the fixed-volume formulas are an off-shell interpolation whose naive volume-uniform reading is refuted), microscopic membership in $\mathcal{S}_W$, and the limit-order control (ML6).

Spin- s slope. For the spin- s ferromagnet the same contact algebra (non-symmetrized free extension plus a double-occupancy channel) gives the closed-form two-magnon amplitude with soft phase slope $1/s$, hard dependence again cancelling; two independent derivations agree to 10^{-15} . This exact computation is the analytic half of the Conjecture B falsifier; it has not yet passed through the verification loop and is counted as evidence, not theorem.

Corner B: flux identity, memory law, Fano reduction

Ground truth: `theory/memory-quantization.md` (rows M-flux PROVED; M-quant PROVED conditional on the hypothesis D18 = H-AD; Mq-AD3 PROVED conditional on the sector-reduction hypothesis Mq-E), `theory/memory-quantization-general.md` (row M-quant-G, PROVED as a conditional implication), `theory/corner-b-draft.md` (rows K1–K3, B3 PROVED; K4 CONJECTURE; row M REFUTED).

Flux identity (exact). For any state, any window $W = [a, b]$, and any finite times, $\frac{d}{dt}\mathfrak{X}_W$ equals $\frac{1}{2s}$ times the difference of the two boundary-bond currents; integrating, $\delta x = \frac{1}{2s}[\tilde{j}_{a-1|a}(0) - \tilde{j}_{b|b+1}(0)]$, the zero-frequency weight of the physical boundary current. No kink, spectral, or asymptotic hypothesis enters. (An earlier reading of this identity through virtual bond data was deleted during verification; the current here is physical.)

Hypothesis H-AD. Wave operators $W_{\pm}$ exist for the selected kink-magnon vector and are complete on its sector; the outgoing space splits into reflected and transmitted channels with definite relative charges $q_{\text{in}} = q_L = -1$, $q_T = +1$; local observables decay on the moving legs; and limits are taken in the stated order (infinite volume, then time, then window). For Theorem 2 in the general register these charges, the density jump $2s$, and summable charge deviations from the two vacua (an ℓ^1 kink class) are the only inputs: conservation of the regularized charge $2s(\mathfrak{X}_W - c)$ plus leg bookkeeping gives $2s\delta x + (q_{\text{out}} - q_{\text{in}}) = 0$ per channel, hence Eq. (2); the quantum for a charge- ν transfer is $-\nu/2s$. The structural remark in the text is the shard’s: covariance forces

$\omega_{g\cdot\alpha}(\xi) = \omega_{\alpha}(\text{Ad}_{g^{-1}}\xi)$, so if the broken element commutes with the unbroken direction ($\text{Ad}_{g^{-1}}\xi = \xi$, as for any abelian G) the two vacua carry *equal* densities and no such kink pair exists; the required inversion is supplied by a Weyl-type element, as the spin flip in XXZ. Nothing in compactness, injectivity, or finite range implies H-AD itself; it is proved here only where stated next.

Sector reduction and the Fano level. In the XXZ chain at $s = \frac{1}{2}$ the incoming component with at most three domain walls, under the enumeration hypothesis Mq-E (verified explicitly at $N = 14$; all-volume form open), is unitarily a two-sided tight-binding chain with one side-coupled level—the flat kink state, whose bondwise annihilation by the exact kink Hamiltonian is proved. Kato–Rosenblum and a Feshbach analysis then give H-AD for this reduced dynamics: trace-class perturbation, complete wave operators, no singular-continuous spectrum. Eliminating the side level yields $t(k) = [1 + iJ^2/4\omega v]^{-1}$ and Eq. (3). Leakage of the full chain out of the reduced sector is measured at $O(\Delta^{-2})$ (probabilities 0.10/0.03/0.008 at $\Delta = 2/4/8$); it affects $T(k)$, not the conservation laws. Full-chain H-AD and the universality of the soft-zero coefficient remain open and are tracked as such in the claims ledger.

Refutation of the naive dictionary. The brief’s Conjecture M (“ δx is the DC limit of the soft factor”) fails on two counts: the channel quantum is independent of the scattering phase once the channels exist, and a phaseless purely transmitting wall still displaces maximally. The refuted row is retained in the ledger with its counterexamples; the surviving pair is the flux identity plus Eq. (2).

Label rigidity. Under stationary vacua and finite-range (Lieb–Robinson) dynamics the factorized vacuum-pair boundary label of the kink sector is preserved at all finite times, so δx is motion within a fixed sector, not a change of sector. The thermodynamic uniqueness/flatness of the kink family (a $\mathbb{Z}$ -torsor reading) remains a conjecture with finite-volume evidence.

The SPT rigidity dichotomy

Ground truth: `theory/spt-rebuild.md` (rows SPT-B-mult, SPT-B’, SPT-E-AKLT, SPT-E’, SPT-T’, SPT-D’, SPT-M’ all PROVED with the hypotheses below explicit; SPT-M’-dyn CONJECTURE; the earlier pointwise bulk-blindness claim and the all-orders edge no-go both REFUTED and retained as negative rows).

Bulk (deformable). A closed on-site insertion leaves $V(g) \otimes \bar{V}(g)$: the multipliers cancel exactly, for every lift, so no closed-bulk amplitude carries a projective anomaly. This does not make bulk data blind to the phase—adjoint representation content distinguishes Pauli-projective from scalar conjugation—but it is deformable: along any common-gap symmetric path with

continuous external data, every normalized bulk coefficient is continuous. On the anisotropic AKLT family ($A_b^x \propto X$, $A_b^y \propto Y$, $A_b^z \propto Z$) the soft-charge coefficient is $C_{\text{bulk}}(b) = b^2/4(1-b^2)$: $\frac{1}{8}$ at the AKLT point, 0.2402 at $b = 0.7$, same phase, same $[\omega]$.

Edge (rigid). On the fixed χ -dimensional Schmidt register of a half-chain, the compensated endpoint residue of a symmetry string is $V(g)$ itself, and the Hermitian charge residue is the centered generator $Q_{\text{edge}} = -iX^\circ$, lift-gauge invariant, with spectrum in the shifted lattice $q_\omega^\circ + \mathbb{Z}$ and every irreducible summand of dimension $\geq d_\omega$ (the minimal ω -projective dimension; > 1 whenever $[\omega] \neq 0$). For the AKLT family the finite-window residue is exactly $-\frac{1}{2}[1 - (2b^2 - 1)^L]Z$, with limit $-Z/2$ independent of b ; the trivial comparison tensor gives 0. The half/integer offset is pinned by the global group ($O(2)$ here, or the $SU(2)$ cover); a bare $U(1)$ does not protect it.

Edge memory. Given H-split, an edge-resolved H-AD, charge conservation, and definite channel charges, $\Delta Q_{\text{edge}} = -(Q_{\text{bulk,out}} - Q_{\text{bulk,in}})$, with channel outcomes in $\mathbb{Z}$ (for the AKLT doublet, in $\{-1, 0, 1\}$)—the same bookkeeping as Theorem 2 with edge charge in place of wall position. Protection is capacity protection: the projective multiplet cannot be split symmetrically. That the specific open AKLT parent Hamiltonian with boundary coupling $P_{0,1}^{(S=2)}$ has a nonzero edge-changing reflection amplitude on an open momentum interval is conjecture SPT-M'-dyn; the deciding half-chain scattering computation, with pre-registered acceptance gates, is specified in the shard. A null result there refutes that conjecture only, not the registered endpoint theorems.

Numerical certificates

Every theorem above carries an independent machine-checked certificate under `theory/checks/` (nine checkers, all passing, each with documented failing “red” mutations), and the quantitative claims of the text trace to committed data files: exact-diagonalization form-factor residuals $\leq 1.6 \times 10^{-14}$ and coefficient fits to 2.2×10^{-10} (`soft_current_recon_check.py`); two-magnon completeness on rings up to 26 sites (`ml2_completeness_check.py`); the flux identity to 3.4×10^{-16} (`mquant_check.py`); the wavepacket memory scan (`memory-scan-1.json`: $N = 160$, sparse kink-sector propagation, trapped weight $< 10^{-6}$, $|\delta x + 2\langle N_T \rangle| \leq 0.004$ sites); the Fano cross-check (0.9–5.8% over $\Delta \in \{1.5, 2, 3, 6, 12\}$, pre-registered 8% criterion); the spin- s falsifier and its independent cross-check (`spin1-bc-falsifier.json`, `spin1-bc-crosscheck.json`: ansatz-free ring spectra to $N = 480$ plus real-time collisions, slope deviations from $1/s$ below 0.5% extrapolated; wavepacket memory ratios within 1% at clean kinematics); and

the SPT deciding computation (`spt_rebuild_check.py`: bulk coefficient and edge residue to stated tolerances, sign and phase-gauge mutants failing). Figures 1–3 are generated from these committed files by `paper/figures/make_figures.py`; no number is hand-entered.

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